CHANGE OF RESIDUE FIELD FOR A VERSAL FORMAL OBJECT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-defos.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the construction of S as the adic completion of a polynomial ring over R together with its category and formal smoothness properties, the reduction to a smoothness statement via the change of residue field lemma, and the verification of the lifting property on morphism sets.

Editorial note. The label lemma-change-of-field-versal-ring names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source formal-defos.tex –label lemma-change-of-field-versal-ring –start 7692 –end 7753 –local-id 28017 returned RESULT=OK (L28017 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-formal-defos/candidates-halfA.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Smoothness of the transformation induced by a change of residue field. Same chapter, the lemma that a smooth transformation stays smooth after the change of residue field construction. Supplies the smoothness of the induced transformation from the localized functor to the changed functor.
- Formal smoothness of a power series over a Noetherian ring in the adic topology. More on Algebra, the lemmas identifying formal smoothness of a ring map with the lifting property and its compatibility with completion. Supplies the formal smoothness of R -> S in the m_S-adic topology and the existence of the lift in the last step.
- Membership of the completion in the category of Artinian quotients. Algebra, the lemma that the completion of a Noetherian local ring is again Noetherian and has the expected residue field. Supplies the membership of S in the category over l.

Source reference. Original lemma 0316: sources/stacks-chow/tags.txt, source lines 3973–3973. Prerequisite pointer invoked at proof line(s) 7711; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:formal-defos.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 07WD: sources/stacks-formal-defos/pinned-original-source.tex, source lines 7674–7674. Prerequisite pointer invoked at proof line(s) 7722; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 06SM: sources/stacks-formal-defos/pinned-original-source.tex, source lines 2598–2598. Prerequisite pointer invoked at proof line(s) 7717; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 07EC: sources/stacks-chow/tags.txt, source lines 9442–9442. Prerequisite pointer invoked at proof line(s) 7714; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 07ED: sources/stacks-chow/tags.txt, source lines 9443–9443. Prerequisite pointer invoked at proof line(s) 7715; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 07NJ: sources/stacks-chow/tags.txt, source lines 9764–9764. Prerequisite pointer invoked at proof line(s) 7752; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 07W8: sources/stacks-formal-defos/pinned-original-source.tex, source lines 7494–7494. Prerequisite pointer invoked at proof line(s) 7694; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *With notation and assumptions as in Situation 07W8. Let ξ be a versal formal object for $\mathcal{F}$ lying over $R \in \mathrm{Ob}(\widehat{\mathcal{C}}_{\Lambda,k})$. Then there exist*

1. *an $S \in \mathrm{Ob}(\widehat{\mathcal{C}}_{\Lambda,l})$ and a local Λ-algebra homomorphism $R \to S$ which is formally smooth in the $\mathfrak{m}_S$-adic topology and induces the given field extension l/k on residue fields, and*
2. *a versal formal object of $\mathcal{F}_{l/k}$ lying over S.*

Proof. Construction of S. Choose a surjection $R[x_1, \ldots, x_n] \to l$ of R-algebras. The kernel is a maximal ideal $\mathfrak{m}$. Set S equal to the $\mathfrak{m}$-adic completion of the Noetherian ring $R[x_1, \ldots, x_n]$. Then S is in $\widehat{\mathcal{C}}_{\Lambda,l}$ by Algebra, Lemma 0316. The map $R \to S$ is formally smooth in the $\mathfrak{m}_S$-adic topology by More on Algebra, Lemmas 07EC and 07ED and the fact that $R \to R[x_1, \ldots, x_n]$ is formally smooth. (Compare with the proof Lemma 06SM.)

Since ξ is versal, the transformation $\underline{\xi} : \underline{R}|_{\mathcal{C}_{\Lambda,k}} \to \mathcal{F}$ is smooth. By Lemma 07WD the induced map

$$(\underline{R}|_{\mathcal{C}_{\Lambda,k}})_{l/k} \longrightarrow \mathcal{F}_{l/k}$$

is smooth. Thus it suffices to construct a smooth morphism $\underline{S}|_{\mathcal{C}_{\Lambda,l}} \to (\underline{R}|_{\mathcal{C}_{\Lambda,k}})_{l/k}$. To give such a map means for every object B of $\mathcal{C}_{\Lambda,l}$ a map of sets

$$\mathrm{Mor}_{\widehat{\mathcal{C}}_{\Lambda,l}}(S, B) \longrightarrow \mathrm{Mor}_{\widehat{\mathcal{C}}_{\Lambda,k}}(R, B \times_l k)$$

functorial in B. Given an element $\varphi : S \to B$ on the left hand side we send it to the composition $R \to S \to B$ whose image is contained in the sub Λ-algebra $B \times_l k$.

Smoothness of the map means that given a surjection $B' \to B$ and a commutative diagram

$$\begin{array}{ccccc} S & \longrightarrow & B & = & B \\ \uparrow & & \uparrow & & \uparrow \\ R & \longrightarrow & B' \times_l k & \longrightarrow & B' \end{array}$$

we have to find a ring map $S \to B'$ fitting into the outer rectangle. The existence of this map is guaranteed as we chose $R \to S$ to be formally smooth in the $\mathfrak{m}_S$-adic topology, see More on Algebra, Lemma 07NJ. $\square$